

DATASET APPROVAL PROPOSAL

EU Campsite Demand Forecasting

1. The Dataset

Source: Eurostat Tourism Statistics; table tour_occ_nim (Nights spent at tourist accommodation establishments, monthly). Filtered to Camping Grounds, RV Parks & Trailer Parks (CAMP) across EU-27 countries, 2010 – present. The resulting dataset has approximately 7,500 – 9,000 rows and features including nights_spent (numeric), geo / country code (categorical), year, and month. Freely available under CC BY 4.0; no personal data. URL: https://ec.europa.eu/eurostat/databrowser/view/tour_occ_nim

2. Classification Question

Can monthly and country-level features predict whether campsite demand will be Low, Medium, or High? The target (demand_level) is derived from nights_spent using country-specific tertile thresholds computed on the training split only, producing three ordered classes that reflect real planning categories.

3. Regression Question

Can seasonal, geographic, and lagged features predict the actual number of nights spent at campsites for a given country-month? The target is the existing nights_spent column (continuous). Lag features (previous month, same month prior year) are derived from past periods only, ensuring no temporal data leakage.

4. Why It Is Interesting

Campsite demand is strongly seasonal and highly sensitive to external shocks, the COVID-19 collapse (2020) and sharp recovery (2021–22) are clearly visible in the data. A model that classifies demand level and forecasts total nights spent gives campsite operators and regional tourism agencies an early-warning signal for capacity planning and investment, all from a single freely available public dataset.

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